

Ashwin Sudarshan

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EDUCATION

University of Maryland

Master of Science in Mechanical Engineering, GPA 3.7

College Park, MD

Aug. 2023 – May 2025

RV College of Engineering

Bachelor's in Mechanical Engineering

Bangalore, India

Aug. 2018 – Aug. 2022

TECHNICAL SKILLS

CAD: Creo, SolidWorks, CATIA V5, Inventor, AutoCAD, GD&T

FEA: ANSYS, ABAQUS, Thermal/Structural FEA

Test & Failure Analysis: Thermal cycling, HAST, Mechanical Shock; SEM/EDS, Real-Time Continuity Monitoring

Data & Automation: Python, MATLAB/Simulink, Power BI, Microsoft Excel, Machine Learning, Artificial Intelligence

Quality & Manufacturing: DFM/DFA, FMEA, SPC, Gauge R&R, Lean/Six Sigma, Root Cause Analysis

Certifications: Six Sigma Black Belt, Agile (Atlassian), Bosch-Certified in 8D

WORK EXPERIENCE

Mechanical Engineer (Research Assistant)

Feb. 2025 – present

University of Maryland

College Park, MD

- Designed an FR-4 test board using Altium CircuitMaker, including a six to eight layer stack, daisy-chain routing, test points, and encapsulant thickness targets, creating a consistent baseline to compare package-level failure modes.
- Selected and specified daisy-chain dummy dies with the supplier and mapped pad nets; decided on wedge bonding and defined pad breakout to the PCB, enabling reliable continuity monitoring on the test boards.
- Built an accelerated-test plan using JEDEC standards to set thermal cycling from – 40 to 125 °C (10-minute dwells, under 20 °C/min, up to 1,000 cycles), temp humidity load at 85 °C and 85% RH for 500 hours, and 1,500 g mechanical shock test; defined failure as an open or a 20 percent resistance rise.

Mechanical Engineering Intern

Jun. 2024 – Aug. 2024

Novelis

Oswego, NY

- Re-engineered burner changeout system in Autodesk Inventor, validated in field trials; cut maintenance time by 35%, saved 150 tech-hours/year, and projected \$30K/year cost savings.
- Standardized the new burner setup procedures with zero safety incidents during adoption.
- Led volumetric analysis on three industrial furnaces; corrected weight-to-volume conversion in Excel and recalibrated radar-level sensors, averting four overfills per year.

Mechanical Design Engineer

Apr. 2022 – Jul. 2023

Bosch

Bangalore, India

- Redesigned the external gear-pump end cover in Solidworks, removing non-critical material while preserving stiffness, which reduced part mass and cut material cost by about \$0.25 per unit.
- Validated the redesigned cover with an Abaqus model using 230-280 bar internal pressure, 26 kN M12 bolt pretension, and housing/bushing contacts; after mesh refinement varying from 33k to 576k elements, the maximum principal stress at the joint converged to 243-247 MPa at 280 bar with stable deformation.
- A 400-bar burst test matched the model, with crack initiating at the predicted hotspot and propagating perpendicular to the principal-stress direction, leading to the adoption of the converged, validated model as the design baseline.
- Developed Power BI dashboards tracking 45 quality metrics across APAC; identified three high-impact failure modes, helping cut defect rates by up to 15%.
- Coordinated with 15 suppliers quarterly to reinforce design/quality standards, reducing delays by 8% and compliance issues by 35%.

PROJECTS

Chassis Design Engineer | Team Helios Racing

2019 – 2021

- Led chassis and braking system integration for ATV prototypes using CATIA V5, HyperMesh, and ANSYS; hands-on fabrication and testing; contributed to team performance at Formula SAE India and Baja SAE Arizona (2020).

F1-Inspired Brake-by-Wire | Indian Institute of Science

2021 – 2022

- Co-developed a brake-by-wire prototype; validated concepts with 45+ static and thermal FEA simulations; created parametric CAD in SolidWorks and led weekly cross-functional design reviews.

F-1 OPT through 2028; no sponsorship required for 3 years. Willing to relocate anywhere in the US and start immediately.